

UNIVERSITY OF BRISTOL

School of Mathematics

FINANCIAL MATHEMATICS 3

MATH35400

(Paper code MATH-35400)

MAY 2024 2 hours 30 minutes

This paper contains 4 questions. All answers will be used for assessment.

- Calculators of an approved type (permissible for A-Level examinations) are permitted.
- Candidates may bring one hand-written sheet of A4 notes written double-sided into the examination. Candidates must insert this into their answer booklet(s) for collection at the end of the examination.

You may find the following results useful:

- *Itô's lemma* for the derivative of a function of an Itô process with stochastic differential $dX_t = b_t dt + \sigma_t dW_t$:

$$df = \left(f_t + b_t f_x + \frac{1}{2} \sigma_t^2 f_{xx} \right) dt + \sigma_t f_x dW_t.$$

Do not turn over until instructed.

1. **(25 marks)** Consider a single-period model with one risky security and a finite probability space $\Omega = \{\omega_1, \omega_2, \omega_3\}$. For each state $\omega \in \Omega$ and time point $t = 0, 1$ let $S_t(\omega)$ denote the price of the security at time point t in state ω . We assume that there is a bank account with interest rate $r = 1/9$ and that the price process is given by the following table:

S_0	$S_1(\omega_1)$	$S_1(\omega_2)$	$S_1(\omega_3)$
45	60	40	30

- (a) **(5 marks)** Specify the value and gains processes V_t and G and their discounted versions V_t^* and G^* for an arbitrary trading strategy $H = (H_0, H_1)$ and for $\omega = \omega_1, \omega_2, \omega_3$ in this single-period model.
- (b) **(6 marks)** Determine all risk-neutral probability measures for this model.
- (c) **(3 marks)** Determine all attainable contingency claims $X = (X_1, X_2, X_3)$ for this model.
- (d) **(3 marks)** Consider European call options on the risky security with strike price $K \geq 0$. Find all possible strike prices such that the call option is attainable and comment on the usefulness of an option with these strike prices for a potential buyer of the option.
- (e) **(2 marks)** State the definition of an arbitrage opportunity. Are there arbitrage opportunities in this market?
- (f) **(6 marks)** A trading strategy H is said to be *dominant* if there exists another trading strategy G such that $V_0^H = V_0^G$ and $V_1^H(\omega) > V_1^G(\omega)$ for all $\omega \in \Omega$. In the model above either prove that there exists no dominant trading strategy or calculate a dominant trading strategy.

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2. **(25 marks)** Consider a two period model with constant interest rate r in both periods and one risky stock whose price process is given in the table below.

ω_k	$t = 0$	$t = 1$	$t = 2$
ω_1	$S_0 = 5$	$S_1 = 8$	$S_2 = 10$
ω_2	$S_0 = 5$	$S_1 = 8$	$S_2 = 6$
ω_3	$S_0 = 5$	$S_1 = 3$	$S_2 = 6$
ω_4	$S_0 = 5$	$S_1 = 3$	$S_2 = 2$

- (a) **(2 mark)** State the difference between a European call option and an American call option on the same stock, with the same strike price and expiry date.
- (b) **(6 marks)** By considering conditional probabilities, calculate all martingale measures for the model above. Your answer should be written in terms of r . In particular, show that, when $r = 0$,

$$\mathbb{Q}(\{\omega_1\}) = 2/10, \quad \mathbb{Q}(\{\omega_2\}) = 2/10, \quad \mathbb{Q}(\{\omega_3\}) = 3/20, \quad \mathbb{Q}(\{\omega_4\}) = 9/20,$$

is the unique martingale measure.

- (c) Consider a European call option on this stock with strike price $K = 7$ and expiry date $T = 2$ and assume that $r = 0$.
- (4 marks)** For each ω_k evaluate the pay-off, then use the risk-neutral valuation principle to find its time zero price.
 - (6 marks)** Find a self-financing trading strategy that replicates the pay-off of this option, and use this to check your answer to the previous part.
- (d) Consider an American claim based on the above stock, which at time $t = 0$ has payoff $Y_0 = 3$ and at time $t \geq 1$ has the payoff process given by $Y_t = (4 - S_t)_+ + (S_t - 4)_+$. Here $(x)_+ = \max(x, 0)$.
- (2 marks)** Calculate the numerical values of the payoff process for each of the states $\omega_1, \dots, \omega_4$.
 - (5 marks)** Determine the optimal strike strategy for this claim when the interest rate is $r = 0$, and state the resulting time 0 value of the claim.

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3. (25 marks)

- (a) Consider a multi-period model with T periods. There is one stock with price process S_t for $t = 0, \dots, T$ and a bank account with a constant interest rate r . We assume that the market is complete.
- (2 marks) What does the completeness of the market imply regarding arbitrage opportunities and martingale measures?
 - (1 mark) Let (Y_t) be the pay-off process for an American call option with strike price K . Give a formula for Y_t in terms of S_t and K .
 - (2 marks) Let τ be a stopping time such that $\tau \leq T$. Without giving any explanation state the time 0 price Π_0 of the American option from (ii) with exercise strategy τ .
 - (5 marks) An investor plans to buy the American call option and use the exercise strategy τ . Show that for any risk-neutral probability measure $\mathbb{Q}$

$$\Pi_0 \leq \mathbb{E}_{\mathbb{Q}}((S_T - K)_+ / (1 + r)^T). \quad (1)$$

Hint: You can use that for any martingale (M_t) , stopping time $\tau \leq T$ and convex function ϕ ,

$$\mathbb{E}(\phi(M_\tau)) \leq \mathbb{E}(\phi(M_T)).$$

- (2 marks) Which financial instrument has the right-hand side of (1) as its price? What does the inequality (1) imply on when to exercise the option?
- (b) Let (W_t) be a standard Brownian motion adapted to a filtration $(\mathcal{F}_t)$.
- (2 marks) Write down Itô's formula for the function $f(t, W_t)$ of a standard Brownian motion W_t , where $f = f(t, x)$ is a function of both t and x .
 - (3 marks) Then use Itô's formula to put the process $X_t = W_t^{2k}$ in standard form $dX_t = \mu_t dt + \sigma_t dW_t$ where k is a positive integer.
 - (3 marks) Use your result from ii. to show that for all integers k

$$\mathbb{E}(W_t^{2k}) = \int_0^t k(2k-1)\mathbb{E}(W_s^{2k-2})ds.$$

Hint: You can use that

$$\mathbb{E}\left(\int_0^t f(W_s)ds\right) = \int_0^t \mathbb{E}(f(W_s))ds$$

for a continuous function f .

- (5 marks) Calculate $\mathbb{E}(W_t^j)$ for $j = 0, \dots, 6$.

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4. **(25 marks)** Let $(W_t)_{t \geq 0}$ be the standard Brownian motion. The Black-Scholes model is a financial market model in continuous time with two assets:

- A riskless bond with constant interest rate r .
- A stock whose price S_t is modelled as

$$S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t).$$

The Black-Scholes formula for the price at time t of a European call option with strike price K and maturing at time T is given by

$$C(t) = S_t \Phi(d_1(S_t, T - t)) - K e^{-r(T-t)} \Phi(d_2(S_t, T - t))$$

where the functions $d_1(s, t)$ and $d_2(s, t)$ are given by

$$d_1(s, t) = \frac{\log(s/K) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$$

$$d_2(s, t) = \frac{\log(s/K) + (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

and Φ is the standard normal cumulative distribution function.

- (a) **(3 marks)** Find the time 0 price of a European call option on a stock when the stock price is 50, the strike price is 60, the risk-free interest rate is 3% per annum, the volatility is 0.3 per annum and the time to maturity is six months. *[You might want to use the table at the end of this exam paper for this question.]*
- (b) **(2 marks)** What is the time 0 price of a European put option with the same strike price as in (a)?
- (c) **(5 marks)** In the above stock price model for S_t , what are the usual names for the parameters μ and σ ? Show that

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

- (d) **(3 marks)** Put $d(S_t^4)$ in standard form $dX_t = \mu_t dt + \sigma_t dW_t$, for some μ_t and σ_t which you should specify.
- (e) **(2 marks)** Let $\mathbb{Q}$ be the equivalent martingale measure. Define the process $\bar{W}_t$ such that it is a standard Brownian motion under $\mathbb{Q}$. Write S_t in terms of $\bar{W}_t$.
- (f) **(7 marks)** Use the risk-neutral valuation principle to calculate the time t price $g(t, S_t)$ of a European option with payoff S_T^4 at time T . *[Hint: For this question you may use that the moment-generating function of a Gaussian random variable X with mean μ and variance σ^2 is $M_X(t) = \exp(\mu t + \sigma^2 t^2/2)$.]*
- (g) **(3 marks)** Show that the function $g(t, x)$ in the previous question satisfies the Black-Scholes-Merton differential equation

$$g_t + rxg_x + \frac{\sigma^2}{2}x^2g_{xx} = rg.$$

End of examination.

Probabilities of a standard normal distribution

The next tables shows probabilities from a standard normal distribution. Each cell contains the value of $\Phi(x_1 + x_2)$ where x_1 and x_2 are the labels of the corresponding row and column.

Φ	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990